GEOPHYSICAL RESEARCH LETTERS, VOL. 7, NO. 7, PAGES 541-544, JULY 1980

PALEOMAGNETIC RESULTS FROM EARLY EOCENE INTRUSIONS, NORTH-CENTRAL MONTANA: IMPLICATIONS FOR NORTH AMERICAN APPARENT POLAR-WANDERING

J. F. Diehl, S. Beske-Diehl

Michigan Technological University, Houghton, MI 49931

M. E. Beck, Jr.

Western Washington University, Bellingham, WA 98225

and B. C. Hearn, Jr.

U. S. Geological Survey, Reston, VA 22092

Abstract. Paleomagnetic poles have been determined from two Early Eocene igneous centers located in the north-central Montana alkalic province. The mean paleomagnetic pole for the Highwood Mountains lies at 81.2°N, 167.3°E (N = 29; α_{95} = 7.1; k = 15.4) while the corresponding pole for the Bearpaw Mountains is located at 80.5°N, 198.4°E (N = 18; α_{95} = 5.8; k = 37.0). Combining these results with studies on Eocene rocks from Wyoming, gives a reference pole located at 81.7°N, 171.2°E with a circle of 95% confidenee of 4.4°. This data, when compared to other early Tertiary results, the Cretaceous stillstand poles and the well grouped mid-Tertiary poles from the continental interior, suggest that for the period 120 to 15 m.y.B.P. apparent polar-wandering for North America occurred as a succession of rapid shifts of the geomagnetic pole separated by periods of stillstand.

Introduction

The early Tertiary (40-65 m.y.B.P.) is one of many intervals of the North American polar-wander path which is not well documented. A knowledge of the exact behavior of the pole relative to North America for this interval is especially important in light of current paleomagnetic research aimed at delineating allochthonous microplates within the western Cordillera and reconstructing their tectonic history. Since any attempt to make quantitative microplate reconstructions with paleomagnetic data depends critically on the existence of well defined reference poles, a precise polar-wander path for the early Tertiary is necessary to unravel the tectonic history of the fragmented pieces that make up the western Cordillera.

Until recently, paleomagnetic data based on a significant number of sampling sites for early Tertiary rocks did not exist [Butler and Taylor, 1978]. Now there are a few good studies that meet this criterion [Shive and Pruss, 1977; Sheriff and Shive, 1980; Butler and Taylor, 1978; Barnes and Butler, 1980]. To add to the data base for this crucial point of time, we began a comprehensive study of the early Tertiary igneous rocks of the north-central Montana alkalic province. This paper presents results from two Early Eocene intrusive centers of the north-central Montana province - the Highwood and Bearpaw Mountains. A companion paper [Jacobson et al., 1980] reports results from the Paleocene intrusives of the province.

Copyright 1980 by the American Geophysical Union.

Geologic Setting

An early Tertiary age for the igneous rocks of the north-central Montana alkalic province is well established. Over 90 K/Ar and fission-track ages suggest that peak igneous activity in the province occurred between 54 and 49 m.y. ago (based on decay constants recommended by the IUGS in 1976) with major intrusions also occurring during the Late Cretaceous and Paleocene, 60-68 m.y. ago [Marvin et al., 1980]. Ages from individual centers suggest that activity in each area lasted approximately 3 to 7 m.y. and that no systematic geographic migration of this igneous activity took place.

The Bearpaw Mountains (54-50 m.y.B.P.) consist of a northern and a southern volcanic field separated by the elongate Bearpaw Mountains arch [Hearn, 1976]. The arch is a composite anticlinal uplift containing numerous stocks, laccoliths, sills, and dikes. These intrusions are composed of biotite pyroxenite, syenite, shonkinite, nepheline syenite, porphyritic latite, monzonite, tinquaite, porphyritic analcime phonolite and carbonatite. The volcanic fields consist of an early series of interlayered mafic phonolite flows, porphyritic latite flows and fragmental deposits, successively overlain by two later series, analcime trachyte flows and mafic analcime phonolite flows. Intrusions also occur within the volcanic fields. Extensive gravity sliding [Reeves, 1946] and related collapse associated with the uplift of the central arch have modified the original structure of the volcanics so that the volcanics now dip toward the arch at an average angle of 35°.

The Highwood Mountains (53-49 m.y.B.P.) are similar to the Bearpaws in that they also consist of both intrusives and volcanics. The earliest volcanic activity produced quartz latite flows, breccias, conglomerates, and tuffs deposited over a terrain of considerable relief [Larson, 1941]. This volcanic pile was then eroded to a hilly terrain before being covered by mafic phonolite flows [Larson, 1941]. With the exception of the amphibole phonolite dikes which appear to be related to the earliest volcanic episode [Buie, 1941], the intrusive rocks of the Highwoods are contemporaneous with the second stage of volcanic activity which produced the mafic phonolite flows [Hurlbut and Griggs, 1939; Burgess, 1941]. Stocks, sills, dikes, and laccoliths were intruded into flat-lying Cretaceous sandstones and shales and/or the associated volcanics [Buie, 1941; Burgess, 1941, Hurlbut and

Paper number 80L0771.
0094-8276/80/0080L-0771$01.00

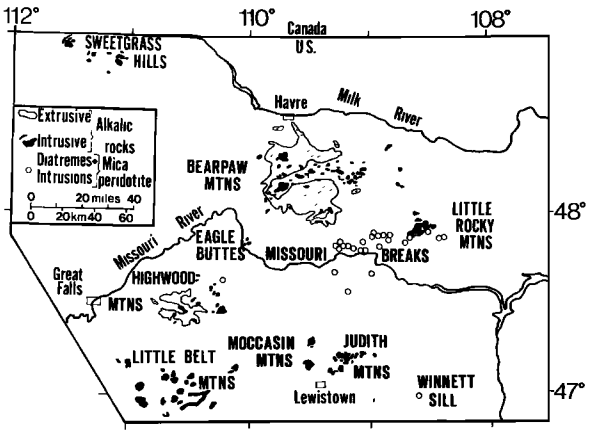


Fig. 1. Index map, modified from Marvin et al. [1980].

Griggs, 1939]. Petrologically, these intrusions consist of shonkinite, syenite, nepheline syenite, biotite pyroxenite, and mafic phonolite [Woods, 1974; Burgess, 1941; Hurlbut and Griggs, 1939; Buie, 1941].

Sampling and Laboratory Procedure

Figure 1 shows the locations of the Bearpaw and Highwood Mountains with respect to the other igneous centers of the north-central Montana province. Sampling of the province took place during the summers of 1977, 1978, and 1979. All samples were collected using a portable gas powered drill and oriented "in situ" using a Brunton compass. Back azimuths and/or sun compass readings were taken to insure against orientation error caused by local magnetic anomalies. Forty-six sites (454 samples) were drilled in the Bearpaws; 23 (243 samples) from the intrusives of the central arch; 12 (118 samples) from the northern volcanic field; and 11 (93 samples) from the southern volcanic field. Because of structural complexities that exist in the volcanic fields due to gravity-sliding, analysis of results from these sites will be deferred to a later paper covering the entire north-central Montana province. In the Highwood Mountains, 39 sites (309 samples) were drilled in a variety of intrusives geographically situated throughout the complex. Sampling was restricted to intrusives where we had good control on the attitude of the adjacent sediments. Dips on the sediment where present were minimal - generally much less than 5° and in most cases horizontal. Because of the problems of questionable initial dip or later tilting, the volcanics were not sampled.

Two samples from each site were subjected to progressive alternating field demagnetization to determine the peak demagnetization field for that site. Optimum demagnetization fields ranged from 10 to 80mT with the median being 20 mT. Magnetic stability was high with most sites having a 95% confidence circle of less than 7 or 8° before and after magnetic cleaning. To insure that all sites yielded results of sufficient accuracy to constitute reliable estimates of the stable ancient geomagnetic field, only sites that had a circle of 95% confidence of less than 15° and a VGP (virtual geomagnetic pole) latitude within 40° of the group mean were used in the final pole calculation. The second criterion eliminates those sites whose magnetization may have been acquired during a reversal or excursion of the geomagnetic field or which have been markedly tilted after magnetization. Five of 23 sites drilled in the Bearpaws failed the reliability criteria; 3 sites had circles of 95% confidence greater than 15° while 2 sites had low latitude VGP's. For the Highwoods, 8 sites failed to meet reliability criteria; 5 sites had low lying VGP's and 3 had circles of 95% confidence greater than 15°. Virtual geomagnetic poles for sites meeting reliability criteria for both the Bearpaw and Highwood Mountains are shown in Figure 2.

Discussion and Results

An important question when dealing with intrusive rocks is the presence or absence of post-magnetization tilt. In the Highwood and Bearpaw Mountains we believe this problem is minimal. The attitude of the sediments near the sampling sites in the Highwoods, as stated before, were much less than 5° and in some cases horizontal. There was no systemmatic direction of observed dips, although Reeves [1946] reports a regional tilt of a degree or less to the northeast. Tilt where present is probably the result of intrusion [Hurlbut and Griggs, 1939]. Therefore, tilting is not a problem in the Highwoods. Control on attitudes in the Bearpaws is less certain. At many of the sites tilting can be shown to be minimal, but at others we are less sure. Since the angular dispersion of the directions from the Bearpaw sites is less that of the Highwoods sites, it is unlikely that the Bearpaw sites have large uncorrected tilts, since if large uncorrected tilts did exist, the dispersion from Bearpaws should be significantly different than the Highwoods. An

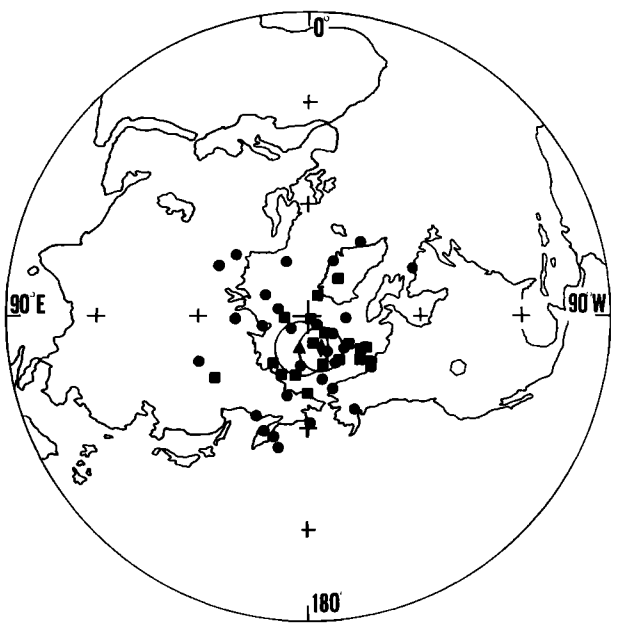


Fig. 2. Polar equal-area projection showing site VGP's for the Highwood (circles) and Bearpaw (squares) Mountains and their mean paleomagnetic poles (triangle and diamond, respectively) with their circles of 95% confidence. Hexagon is sampling location.

19448007, 1980, 7, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL007i007p00541 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Mean directions of remanent magnetization and paleomagnetic pole positions for Early Eocene rocks from the Bearpaw and Highwood Mountains, Montana. Mean pole positions are the vector means of the virtual geomagnetic pole position for each site.

		Mean direction					Mean pole				
	N(#)	D	I	α_{95}	k	δ	Lat.	Long.	α_{95}	k	δ
Bearpaws	18(10)	347.0	69.0	4.3	71.8	9.3	80.5°N	198.4°E	5.8	37.0	13.0
Highwoods	29(0)	347.2	64.4	5.2	27.8	15.1	81.2°N	167.3°E	7.1	15.4	20.4

Explanations: N, number of sites meeting reliability criteria;
(#), number of reversely magnetized sites;
α_{95}, k, statistical parameters [Fisher, 1953];
δ, angular dispersion [Wilson, 1959].

F-test performed on the two populations showed their dispersion to be identical at the 95% confidence level. Since the dip of the sediments along the periphery of the central arch show no asymmetry, regional tilting of the Bearpaws can also be ruled out. If the Bearpaws and the Highwoods are affected by small amounts of post-magnetization tilt it is not of sufficient magnitude to affect the mean directions determined from the two intrusive complexes.

Radiometric ages from the Highwood and Bearpaw Mountains indicate that the duration of igneous activity at each was approximately 4 m.y. and that the magnetically sampled intrusions in the Highwoods may be slightly younger than the early part of the Bearpaw Mountains sequence. Since our sampling scheme covered all the major dated and undated lithologic units, it is reasonable to assume that the calculated poles are indeed paleomagnetic poles and that both represent real points on the apparent polar-wander path for North America. The mean directions and poles for the two areas along with their associated dispersion statistics are listed in Table 1. (For the Highwoods, four sites from the Shonkin Sag laccolith were combined into two composite sites, thereby reducing N to 29). As seen in Figure 2, the mean poles are statistically different from the present rotational axis at the 95% probability level.

Results from this study compare extremely well with those from the Eocene (48 m.y.) volcanics of the Absaroka Mountains, Wyoming [Shive and Pruss, 1977] and the Eocene (44 m.y.) intrusives of the Rattlesnake Hills, Wyoming [Sheriff and Shive, 1980]. The close grouping of these poles is quite remarkable (see Figure 3) considering that these results come from 4 distinct geographic areas and would suggest that studies have truly recorded the Eocene paleomagnetic field. However, the work of [Lovlie and Opdyke, 1974] on felsic and basic intrusives in Virginia, dated at 47 m.y.B.P. gives a pole (87.6°N, 45.0°E; α_{95} = 12.0) that deviates from the above studies and is more characteristic of mid-Tertiary results (see Figure 3). Lovlie and Opdyke did not calculate a pole position from their work, which is based on only thirty-four samples from six sites, one of which contained only two samples. In light of the small number of sites and the divergence of this pole from those of Wyoming and this study, we have not used the Virginia results in our calculations. Thus, we compute a mean Eocene paleomagnetic reference pole for the stable interior of North America using only the results of this study and those from Wyoming. The mean pole position so calculated lies at 81.7°N, 171.2°E with a confidence circle of 4.4°.

Figure 3 shows poles of Cretaceous, early, and middle Tertiary age for the stable interior of North America. Three rather well defined groups are apparent; a Cretaceous group consisting of poles for rock units ranging in age from about 120 to 75 m.y. centered on 67.7°N, 186.0°E (α_{95} = 2.2); an early Tertiary group consisting of poles for rock units 65 to 44 m.y. in age centered on 80.9°N, 180.4°E (α_{95} = 3.4); and a middle Tertiary group from rock units 35 to 15 m.y. centered on 83.8°N, 91.6°E (α_{95} = 5.2). Apparently, between 75 and 65 m.y. and again between 44 and 35 m.y. the pole moved some 13° and 10°, respectively, relative to North America with times of

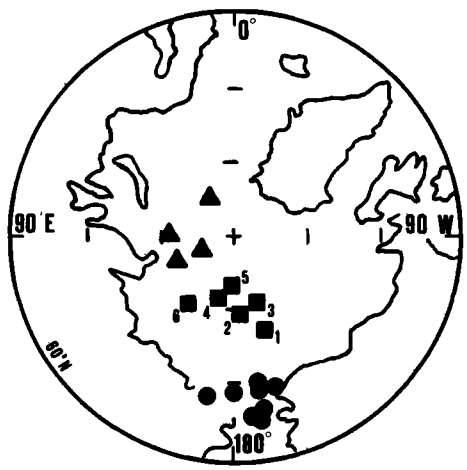


Fig. 3. Cretaceous (circles), early Tertiary (squares), and middle Tertiary (triangles) paleomagnetic poles for North America. Cretaceous poles are those listed in Table 3 of Mankinen [1978]. Data for middle Tertiary pole: Gromme et al. [1972], Gose [1970], Best et al. [1973], Beck et al. [1977], Gilliland et al. [1969], Young and Brennan [1974]. Data for early Tertiary: 1, Gringo Gulch volcanics [Barnes and Butler, 1980]; 2, Paleocene intrusives [Jacobson et al., 1980]; 3, Bearpaw Mtns. (this paper); 4, Highwood Mtns. (this paper); 5, Absaroka volcanics [Shive and Pruss, 1977]; 6, Rattlesnake Hills intrusives [Sheriff and Shive, 1980].

19448007, 1980, 7, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL007i007p00541 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

stillstand (little apparent motion) between these rapid shifts. These periods of rapid shift may correlate with major changes in plate geometry and velocities that occurred at about 80 and 40 m.y.B.P., as suggested by Coney [1972, 1978].

Acknowledgments. Special thanks are owed to K. Koskelin, P. Vance, K. Enzor, R. Schoonover, and K. McClannahan, for direct assistance in the laboratory. K. Koskelin and P. Vance also assisted in the field. This work was supported by NSF grant EAR 78-03652.

References

Barnes, A.E. and R.F. Butler, A paleomagnetic pole from the Gringo Gulch volcanics, Santa Cruz County, Arizona, Geophys. Res. Lettrs., in press, 1980.

Beck, M.E., Jr., S.D. Sheriff, J.F. Diehl, E.A. Hailwood, and P.W. Lipman, Further paleomagnetic results from the San Juan volcanic field of southern Colorado, Earth Plant. Sci. Lettrs., 37, 124-130, 1977.

Best, M.G., R.T. Shuey, C.F. Caskey, and S.K. Grant, Stratigraphic relations of the Needles Range Formation at type localities in southwestern Utah, Bull. Geol. Soc. Amer., 34, 3269-3278, 1973.

Buie, B.F., Igneous rocks of the Highwood Mountains, Montana, Part III. Dikes and related intrusives, Bull. Geol. Soc. Amer., 52, 1753-1808, 1941.

Burgess, C.H., Igneous rocks of the Highwood Mountains, Montana, Part IV. The stocks, Bull. Geol. Soc. Amer., 52, 1809-1828, 1941.

Butler, R.F. and L.H. Taylor, A middle Paleocene paleomagnetic pole from the Nacimiento Formation, San Juan Basin, New Mexico, Geology, 6, 495-498, 1978.

Coney, P.J., Cordilleran tectonics and North American plate motion, Amer. J. Sci., 272, 603-628, 1972.

_____, Mesozoic-Cenozoic Cordilleran plate tectonics, in Smith, R.D. and G.P. Eaton, eds., Cenozoic tectonics and regional geophysics of the western Cordillera, Geol. Soc. Amer. Memoir 152, 33-50, 1978.

Fisher, R., Dispersion on a sphere, Proc. R. Soc., A217, 295-305, 1953.

Gilliland, M.W., H.C. Clark, and J.F. Sutter, Paleomagnetism of the Buck Hills volcanic series, Big Bend, Texas (abs.), Trans. Am. Geophys. Union, 50, 131, 1969.

Gose, W.A., Paleomagnetic studies of Miocene ignimbrites from Nevada, Geophys. J., 20, 241-252, 1970.

Gromme, C.S., E.H. McKee, and M.C. Blake, Paleomagnetic correlation and potassium-argon dating of middle Tertiary ash-flow sheets in the eastern Great Basin, Nevada and Utah, Bull. Geol. Soc. Amer., 83, 1619-1638, 1972.

Hearn, B.C., Jr., Geologic and tectonic maps of the Bearpaw Mountains area, north-central Montana, U.S. Misc. Geol. Inv. Series, Map I-919, 1976.

Hurlbut, C.S., Jr., and D. Griggs, Igneous rocks of the Highwood Mountains, Montana, Part I. The laccoliths, Bull. Geol. Soc. Amer., 50, 1043-1112, 1939.

Jacobson, D., M.E. Beck, Jr., J.F. Diehl, B.C. Hearn, Jr., A Paleocene paleomagnetic pole for North America from alkalic intrusives, central Montana, Geophys. Res. Lettrs., in press, 1980.

Larson, E.S., Igneous rocks of the Highwood Mountains, Montana, Part III. The extrusive rocks, Bull. Geol. Soc. Amer., 52, 1733-1752, 1941.

Lovlie, R., and N.D. Opdyke, Rock magnetism and paleomagnetism of some intrusions from Virginia, J. Geophys. Res., 79, 343-349, 1974.

Mankinen, E.A., Paleomagnetic evidence for a late Cretaceous deformation of the Great Valley sequence, Sacramento Valley, California, J. Res. U.S. Geol. Surv., 6, 383-390, 1978.

Marvin, R.F., B.C. Hearn, Jr., H.H. Mehnert, C.W. Naeser, R.E. Zartman, and D.A. Lindsey, Late Cretaceous - Paleocene - Eocene igneous activity in north-central Montana, Isochron/West, in press, 1980.

Reeves, F., Origin and mechanics of thrust faults adjacent to the Bearpaw Mountains, Bull. Geol. Soc. Amer., 57, 1033-1048, 1946.

Sheriff, S.D., and P.N. Shive, The Rattlesnake Hills of central Wyoming revisited: Further paleomagnetic results, Geophys. Res. Lettrs., in press, 1980.

Shive, P.N., and E.F. Pruss, A paleomagnetic study of basalt flows from the Absaroka Mountains, Wyoming, J. Geophys. Res., 82, 3039-3048, 1977.

Wilson, R.L., Remanent magnetism of the late Secondary and early Tertiary rocks, Phil. Mag., 4, 750-755, 1959.

Woods, M.J., Textural and geochemical features of the Highwood Mountains volcanics, central Montana, Ph.D. Dissertation, University of Montana, Missoula, 1974.

Young, R.A., and W.J. Brennan, Peach Springs tuff: Its bearing on structural evolution of the Colorado Plateau and development of Cenozoic drainage in Mohave County, Arizona, Bull. Geol. Soc. Amer., 85, 83-90, 1974.

(Received April 25, 1980;
accepted May 27, 1980.)

19448007, 1980, 7, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL007i007p00541 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License